University of Petra

Faculty of business

Project title:

[Charging points demand predictions]

By:

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Contents

..... 1
List of figures. 3
Acknowledgements: 4
Abstract: 5
Introduction: 6
Objectives: 6
Data set: 7
Data analysis: 8
Target attribute: 9
Data visualization: 10
Data preparation: 10
Select attributes: 11
Normalization: 11
Correlation: 13
Modeling: 14
Decision tree model: 14
Validation: 15
Tree: 16
Performance vector: 17
Accurecy: 17
Random forest: 18
Process: 18
MODEL: 19
Performance vector: 20
Accuracy: 20
Naïve bayes: 21
K-Means: 23
Data visualization: 26
Conclusion: 28

List of figures.

Figure 1 chart about increas EV usage in jordan (2015-2023) 6
Figure 2: data set 7
Figure 3: import dataset 8
Figure 4: analysis for (Meter value) attribute 8
Figure 5: Analysis for (Stop reason) attribute 9
Figure 6: Targeted attribute (connector type) 9
Figure 7: Data visualization 10
Figure 8: Data preparation 11
Figure 9: selected attribute 11
Figure 10: Narmalization 12
Figure 11: Correlation 13
Figure 12: Modeling process 14
Figure 13: Modeling validation 15
Figure 14: Modeling tree 16
Figure 15: Modeling performance vector 17
Figure 16: Modeling accurecy 17
Figure 17:Random forest process 1 18
Figure 18:Random forest process 2 18
Figure 19:Random forest modeling 19
Figure 20: Random forest predictions 19
Figure 21: Random forest performance 20
Figure 22: Random forest accuracy 20
Figure 23: Naive bayes model 1 21
Figure 24: Naïve bayes model 2. 22
Figure 25: naive bayes result 1 23
Figure 26:Naive bayes result 2 23
Figure 27: K-Means model 24
Figure 28: K-Means clustring result 1 24
Figure 29: K-Means clustring result 2 25
Figure 30: K-Means clustring result 2 25
Figure 31: Power BI result 1 26
Figure 32:Power BI result 2 27
Figure 33:Power BI result 3 27

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I addition, I would like to thank my brother, who was major supporter of the project idea, and I thank my family and friends for the support and all of their contributions during this journey.

Abstract:

This thesis addresses the problems faced by electric cars charging points in Jordan, focusing on the problems, analyzing them, and giving best results that can be obtained. The research objectives include providing the best service to electric car users and alleviating the problems they face, for example, type of connecter, charging period, peek time, and the geographical locations in Jordan most used for charging points.

Introduction:

Recently, electric cars have spread in Jordan and the demand for them has increased. Many companies now manufacture this type of car, and we have noticed the presence of new brands that specialize in electric cars. One of the problems faced by electric car users is the lack of large charging stations to meet their needs. Which led to the desire to discuss this project and put it in your hands, as no solutions to this problem have been found yet.

I show you a graph showing the extent of the increase in demand for electric cars (2015-2023).

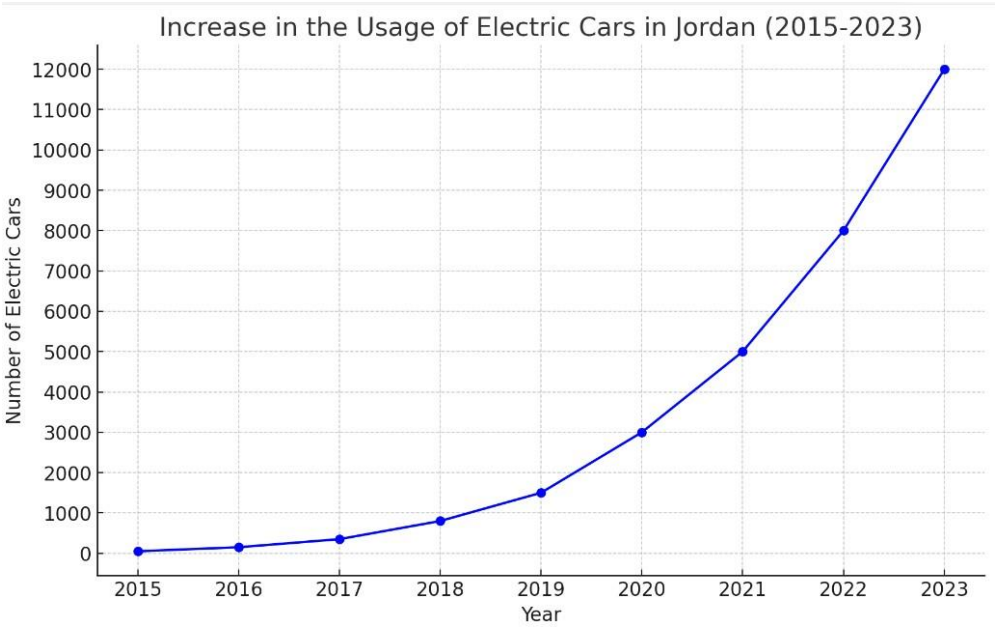
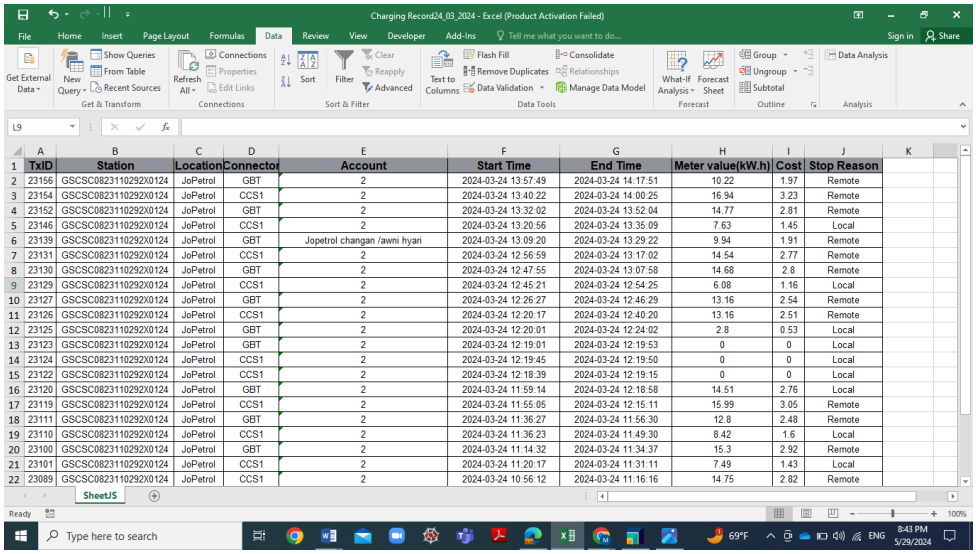


Figure 1 chart about increas EV usage in jordan (2015-2023)

Objectives:

1. Create a predictive model that can correctly forecast to take the best decision.
2. Use the right measure to assess the prediction model success
3. Based on the analysis, offer advice to those responsible

Data set:



TxID	Station	Location	Connector	Account	Start Time	End Time	Meter value(kW.h)	Cost	Stop Reason
23156	GSCSC0823110292X0124	JoPetrol	GBT	2	2024-03-24 13:57:49	2024-03-24 14:17:51	10.22	1.97	Remote
23154	GSCSC0823110292X0124	JoPetrol	CCS1	2	2024-03-24 13:40:22	2024-03-24 14:00:25	16.94	3.23	Remote
23152	GSCSC0823110292X0124	JoPetrol	GBT	2	2024-03-24 13:32:02	2024-03-24 13:52:04	14.77	2.81	Remote
23146	GSCSC0823110292X0124	JoPetrol	CCS1	2	2024-03-24 13:20:56	2024-03-24 13:35:09	7.63	1.45	Local
23139	GSCSC0823110292X0124	JoPetrol	GBT	Jopetrol changan /awni hyari	2024-03-24 13:09:20	2024-03-24 13:29:22	9.94	1.91	Remote
23131	GSCSC0823110292X0124	JoPetrol	CCS1	2	2024-03-24 12:56:59	2024-03-24 13:17:02	14.54	2.77	Remote
23130	GSCSC0823110292X0124	JoPetrol	GBT	2	2024-03-24 12:47:55	2024-03-24 13:07:58	14.68	2.8	Remote
23129	GSCSC0823110292X0124	JoPetrol	CCS1	2	2024-03-24 12:45:21	2024-03-24 12:54:25	6.08	1.16	Local
23127	GSCSC0823110292X0124	JoPetrol	GBT	2	2024-03-24 12:26:27	2024-03-24 12:46:29	13.16	2.54	Remote
23126	GSCSC0823110292X0124	JoPetrol	CCS1	2	2024-03-24 12:20:17	2024-03-24 12:40:20	13.16	2.51	Remote
23125	GSCSC0823110292X0124	JoPetrol	GBT	2	2024-03-24 12:20:01	2024-03-24 12:24:02	2.8	0.53	Local
23123	GSCSC0823110292X0124	JoPetrol	GBT	2	2024-03-24 12:19:01	2024-03-24 12:19:53	0	0	Local
23124	GSCSC0823110292X0124	JoPetrol	CCS1	2	2024-03-24 12:19:45	2024-03-24 12:19:50	0	0	Local
23122	GSCSC0823110292X0124	JoPetrol	CCS1	2	2024-03-24 12:18:39	2024-03-24 12:19:15	0	0	Local
23120	GSCSC0823110292X0124	JoPetrol	GBT	2	2024-03-24 11:59:14	2024-03-24 12:18:58	14.51	2.76	Local
23119	GSCSC0823110292X0124	JoPetrol	CCS1	2	2024-03-24 11:55:05	2024-03-24 12:15:11	15.99	3.05	Remote
23111	GSCSC0823110292X0124	JoPetrol	GBT	2	2024-03-24 11:36:27	2024-03-24 11:56:30	12.8	2.48	Remote
23110	GSCSC0823110292X0124	JoPetrol	CCS1	2	2024-03-24 11:36:23	2024-03-24 11:49:30	8.42	1.6	Local
23100	GSCSC0823110292X0124	JoPetrol	GBT	2	2024-03-24 11:14:32	2024-03-24 11:34:37	15.3	2.92	Remote
23101	GSCSC0823110292X0124	JoPetrol	CCS1	2	2024-03-24 11:20:17	2024-03-24 11:31:11	7.49	1.43	Local
23089	GSCSC0823110292X0124	JoPetrol	CCS1	2	2024-03-24 10:56:12	2024-03-24 11:16:16	14.75	2.82	Remote

Figure 2: data set

The following attributes are included in data set:

1.TXID	Movement serial number
2. Station	Charger serial number
3. Location	The area that used to charge
4. Connector	EV plug type
5. Account	The user count name
6. Start time	EV session starting time
7. End time	EV session ending time
8. Meter value(kW.h)	Kilo wat per hour
9. Cost	Charger session cost (JOD)
10. Stop reason	Charger session stop method

Data analysis:

I brought my Excel file into the process area by dragging it from the repository, where I had initially stored it. Within the operators, I entered "statistics," connected it with the Retrieve Excel File operator, and initiated the process. This enabled me to visualize the distribution through individual bar charts for each column. Additionally, I obtained valuable statistics such as minimum, maximum, average, and standard deviation.

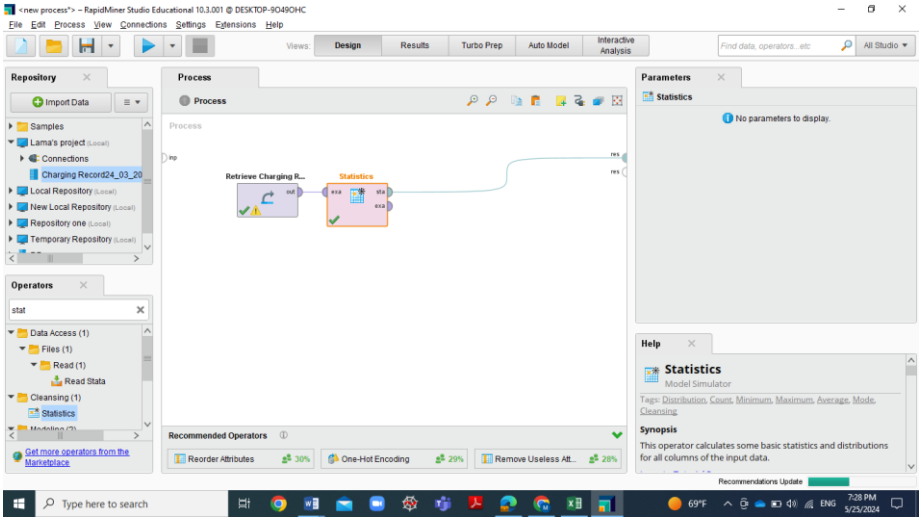


Figure 3: import dataset

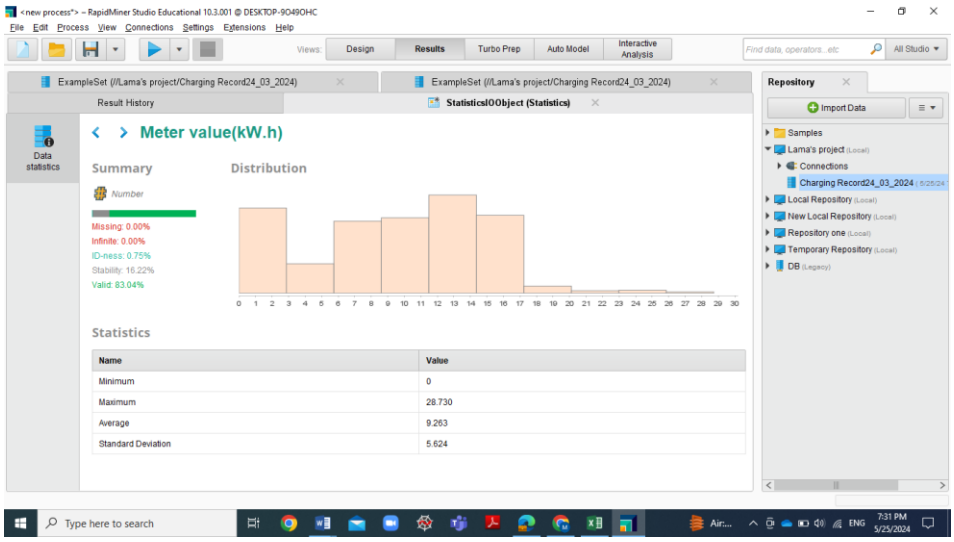


Figure 4: analysis for (Meter value) attribute

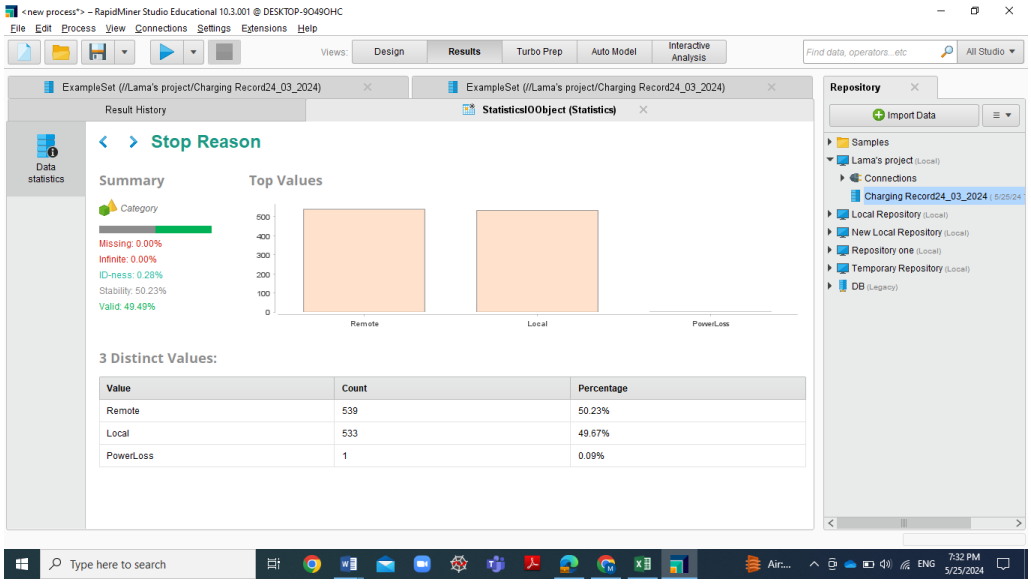


Figure 5: Analysis for (Stop reason) attribute

Target attribute:

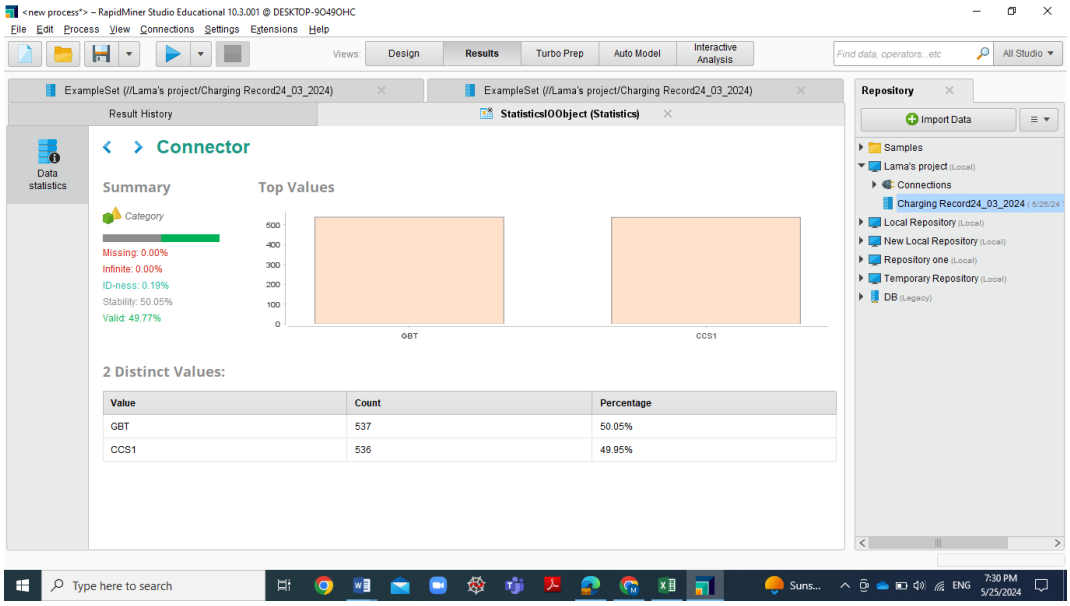


Figure 6: Targeted attribute (connector type)

Now we talking about two types of connecter (GBT, CCS1) they're not equaled but the difference between them less of 1%, So the (GBT) is the most connecter used in our data.

Data visualization:

Following this, I added another Retrieve operator from the repository and executed the process. Subsequently, in the right panel, I navigated to "Visualizations" to create four charts, allowing me to explore the relationships between different columns in my dataset.

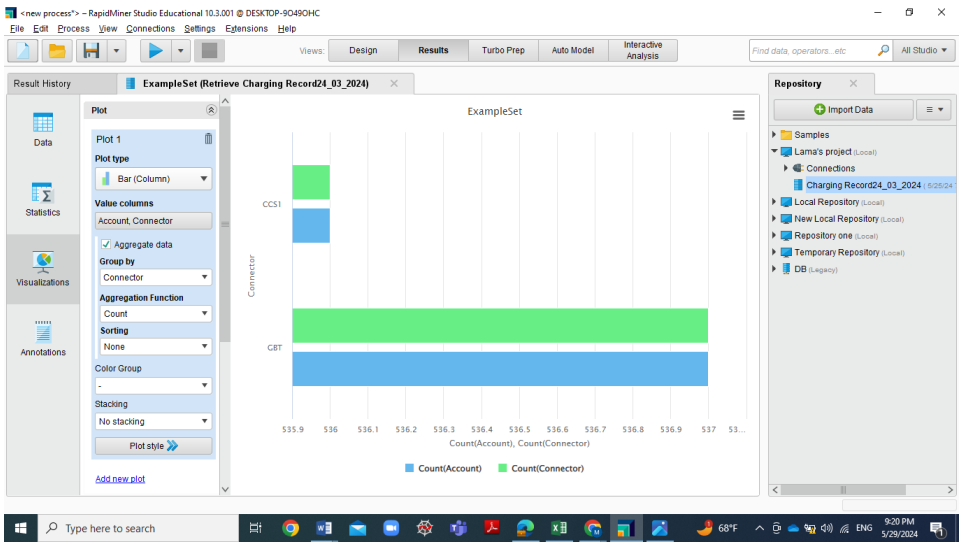


Figure 7: Data visualization

Data preparation:

- First operater we use is “ Retrieve” to load the data into process
- Second operater we use is “select attributes” to select the attributes that we need to apply our model.
- Next attribute we use “Normalize” this operater normalizes by z-transformation
- Correlation matrix: this operater determines correlation between all attributes, and it can produce a weights vector based on these correlations

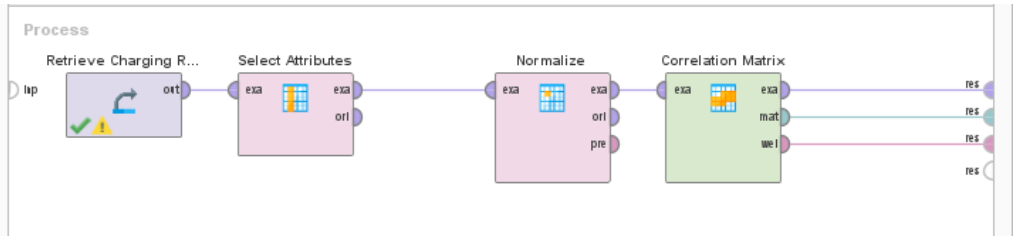


Figure 8: Data preparation

Select attributes:

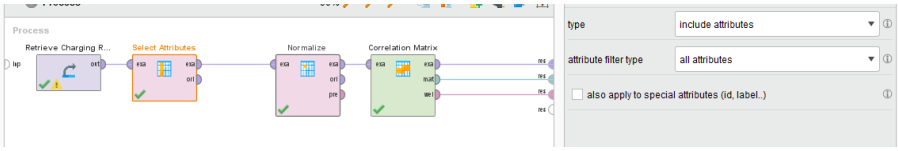


Figure 9: selected attribute

We didn't use the following attributes:

- I selected all the attributes
- We need all the attributes to be selected to have the most the accurate prediction

Normalization:

The goal of normalization is to change the values of numeric columns in the dataset to use a common scale, without distorting differences in the ranges of values or losing information.

We use z-transformation method z-transformation Z score is also called standard score. This score helps to understand the data value.

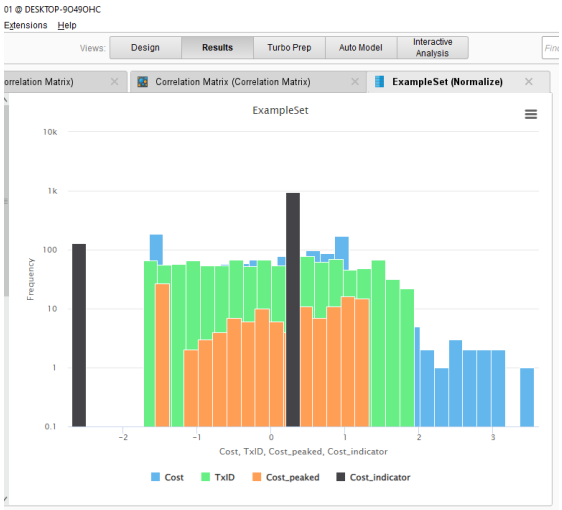
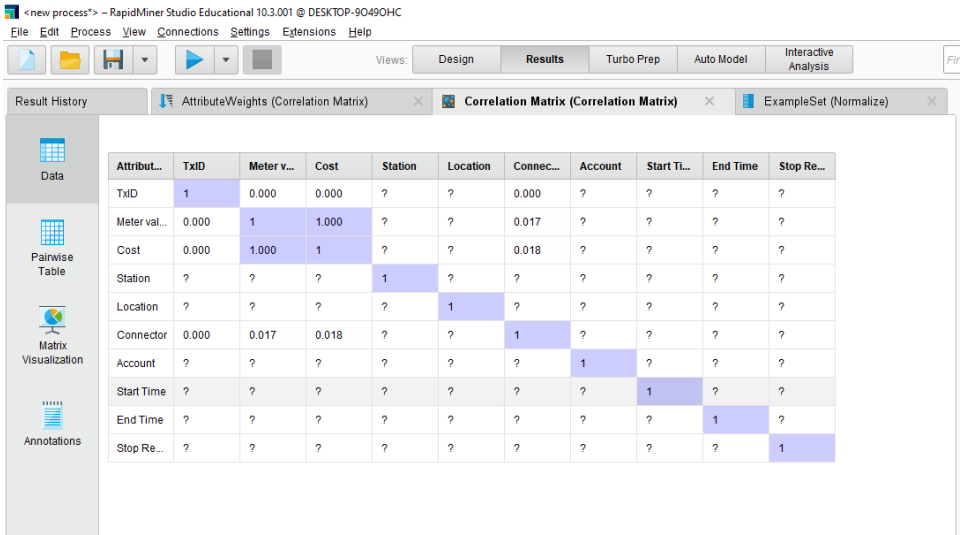


Figure 10:Normaliz charts

Normalizing data helps us to improve data accuracy and enhanced data integrity. It lays the foundation for effective data analysis and decision-making, ensuring that organizations derive meaningful insights from their data assets.

Correlation:

It is a statistical method used to measure the strength of linear relationship between the variables . Correlation analysis calculates the level of change in variables.



Attribut...	TxID	Meter v...	Cost	Station	Location	Connec...	Account	Start Ti...	End Time	Stop Re...
TxID	1	0.000	0.000	?	?	0.000	?	?	?	?
Meter val...	0.000	1	1.000	?	?	0.017	?	?	?	?
Cost	0.000	1.000	1	?	?	0.018	?	?	?	?
Station	?	?	?	1	?	?	?	?	?	?
Location	?	?	?	?	1	?	?	?	?	?
Connector	0.000	0.017	0.018	?	?	1	?	?	?	?
Account	?	?	?	?	?	?	1	?	?	?
Start Time	?	?	?	?	?	?	?	1	?	?
End Time	?	?	?	?	?	?	?	?	1	?
Stop Re...	?	?	?	?	?	?	?	?	?	1

Figure 11: Correlation

- For the attribute (Meter value) we have the cost attribute as its highest correlation.

Modeling:

For our analysis we will use the following models:

- Decision tree model
- Random forest mode
- Naïve Bayes
- K-Means

Decision tree model:

Decision tree model analysis involves making a tree-shaped diagram to chart out a course of action or statistical probability analysis. It used to break down complex problems or branches.

Process:

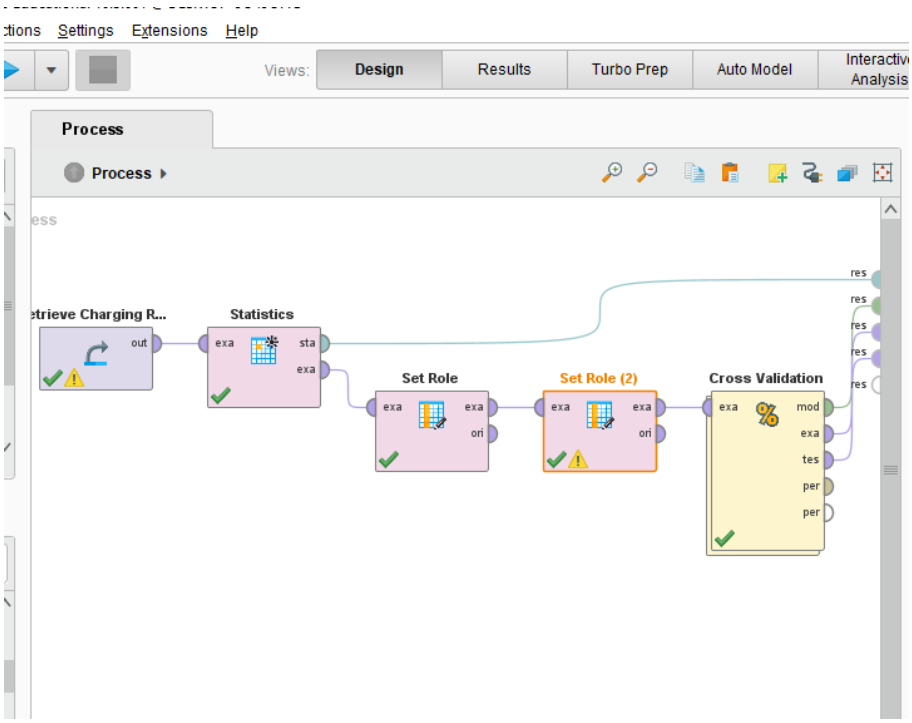


Figure 12: Modeling process

Validation:

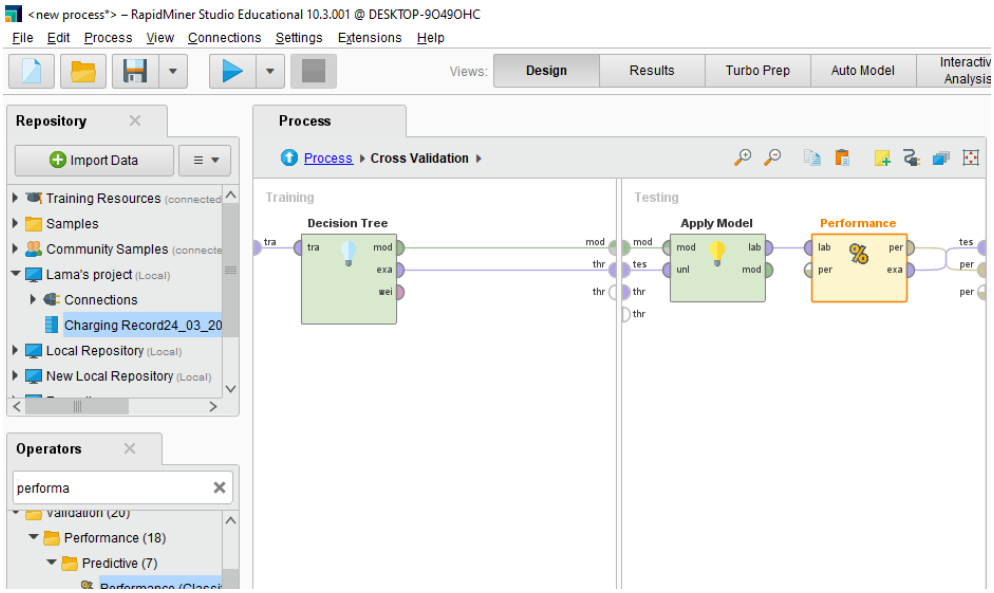


Figure 13: Modeling validation

- We use the set role operator to set (Connector) as label data
- Apply Model operator for applying the subset of the data to trained model
- Perfomance operator for evaluating the performance of model
- The decision tree model to set the parameters as shown in figure

Tree:

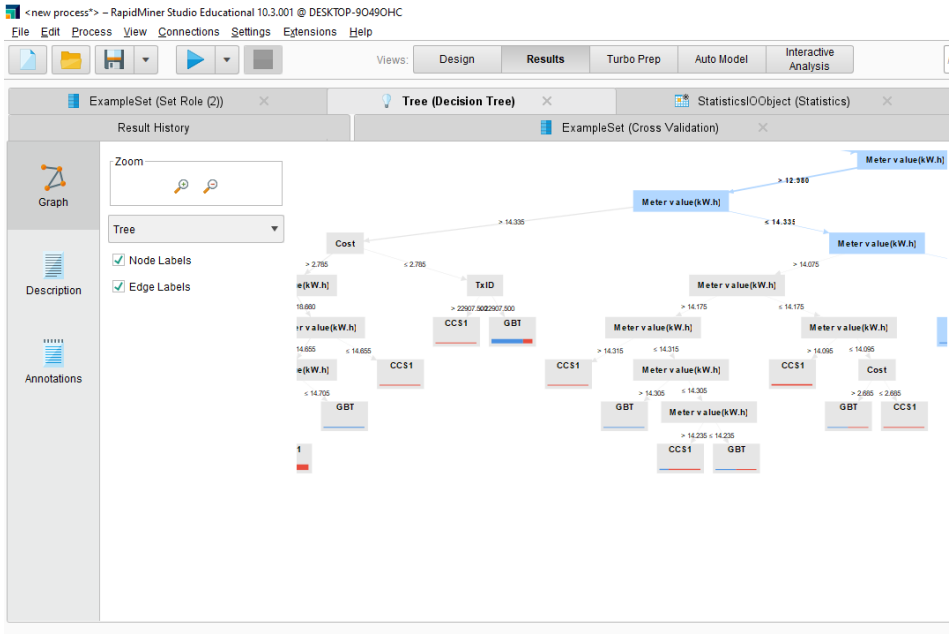


Figure 14: Modeling tree

As shown in the figure the highest prediction for the connector type is the “GBT” is the most used type to charge, So it constituted 50.05%

Performance vector:

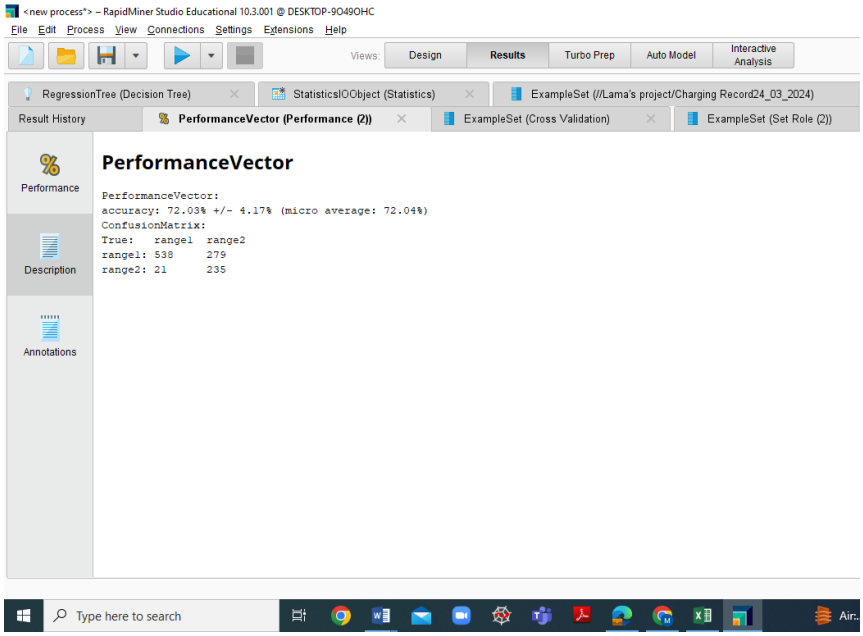


Figure 15: Modeling performance vector

Accurecy:

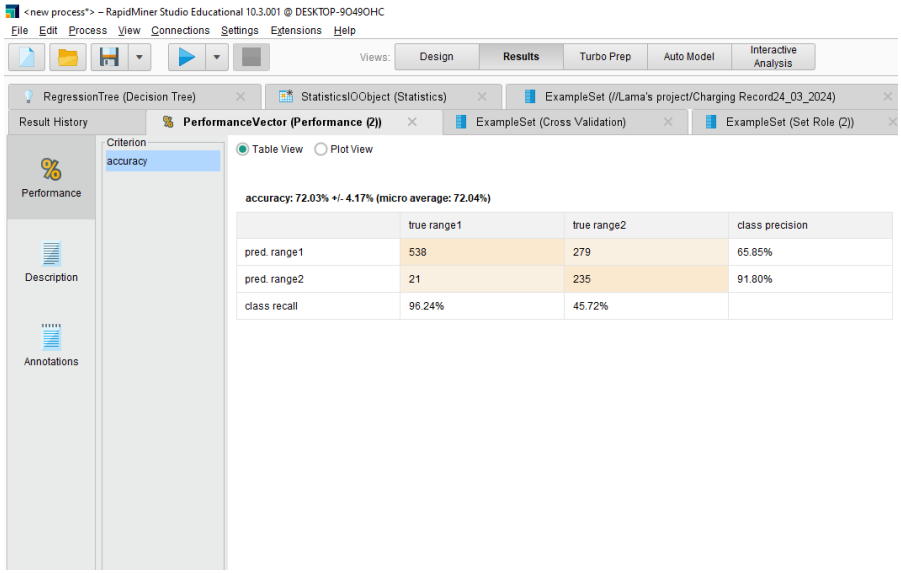


Figure 16: Modeling accurecy

Random forest:

Random forest regression is a supervised learning algorithm that uses to ensemble learning method for regression.

Process:

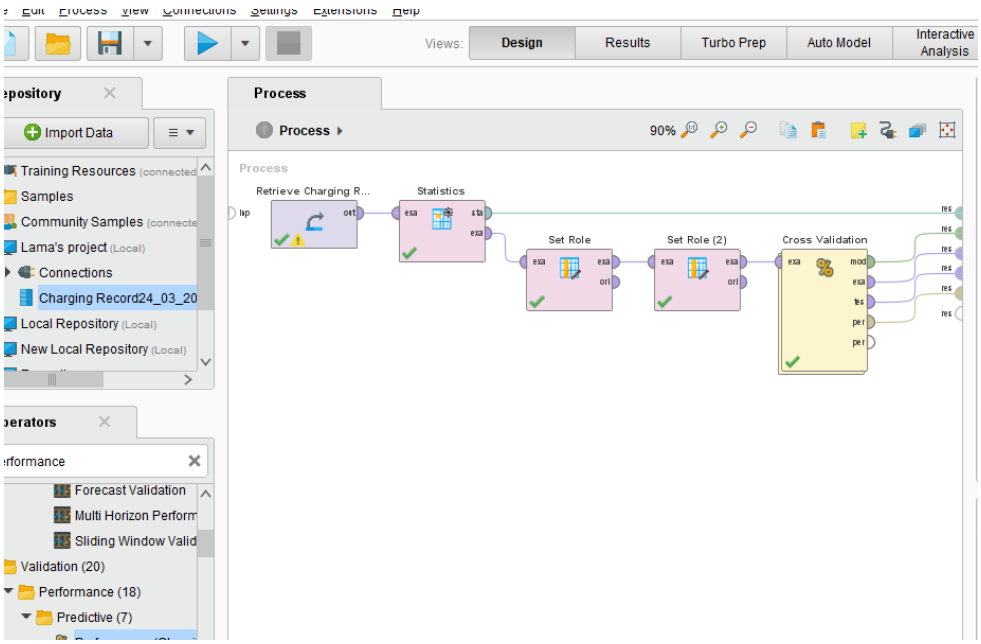


Figure 17:Random forest process 1

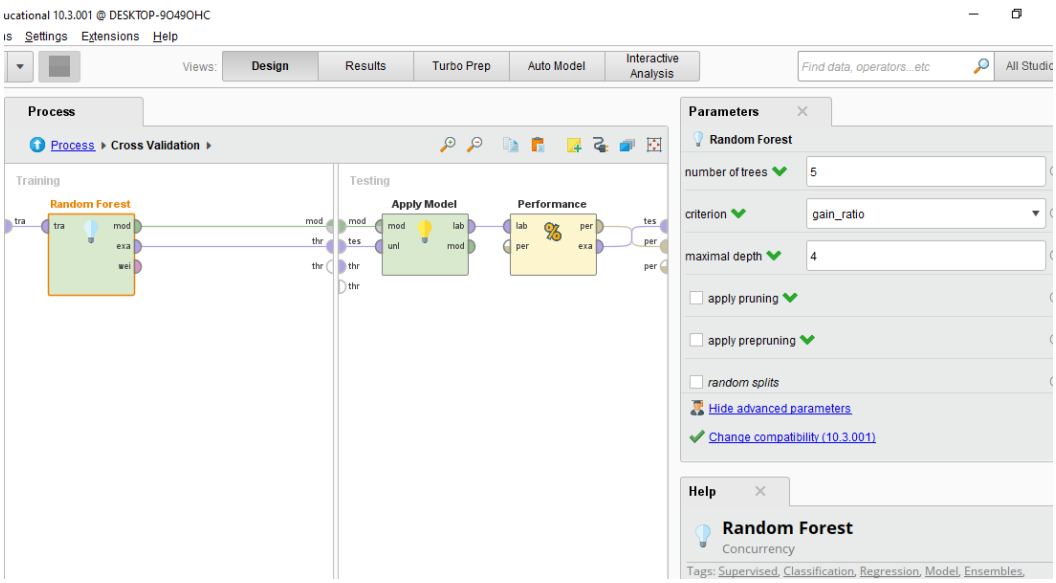


Figure 18:Random forest process 2

- We use the set operator to set “connector” as our label
- Cross validation, the data set is divided into multiple subsets or folds.
- Performance operator for evaluating the performance of model
- The random forest model is to set on the parameters as shown which are the default settings:

1.Number of trees (5)

2. Maximal depth of model (4).

3. Only the least square criterion can be used for numeric label

MODEL:

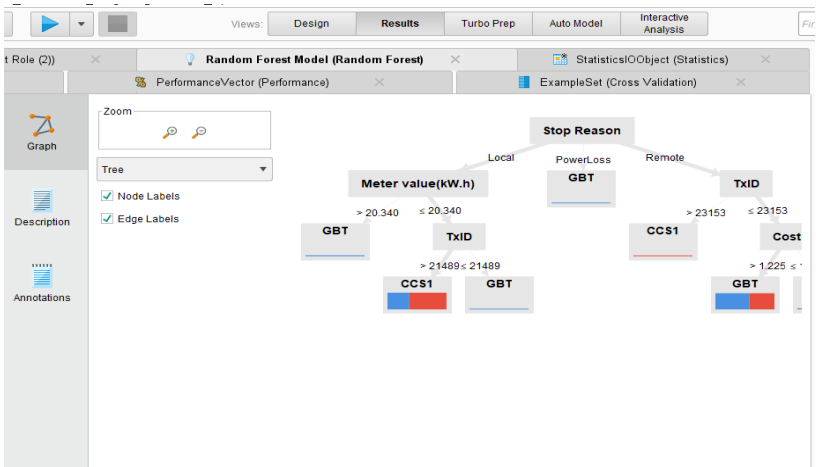


Figure 19:Random forest modeling

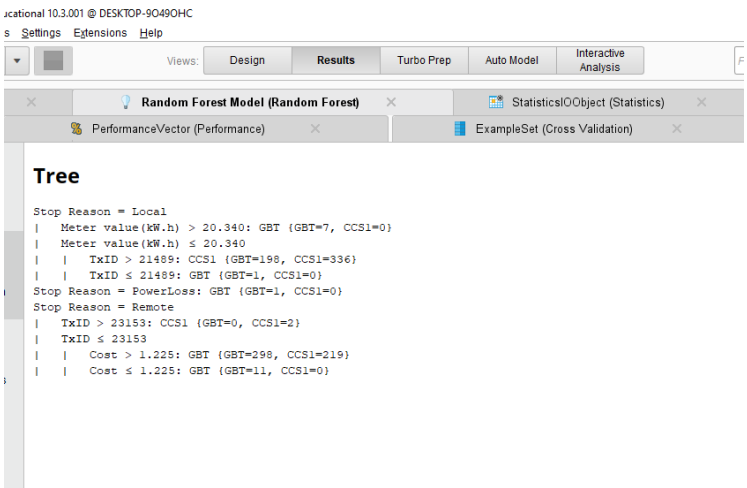


Figure 20: Random forest predictions

Performance vector:

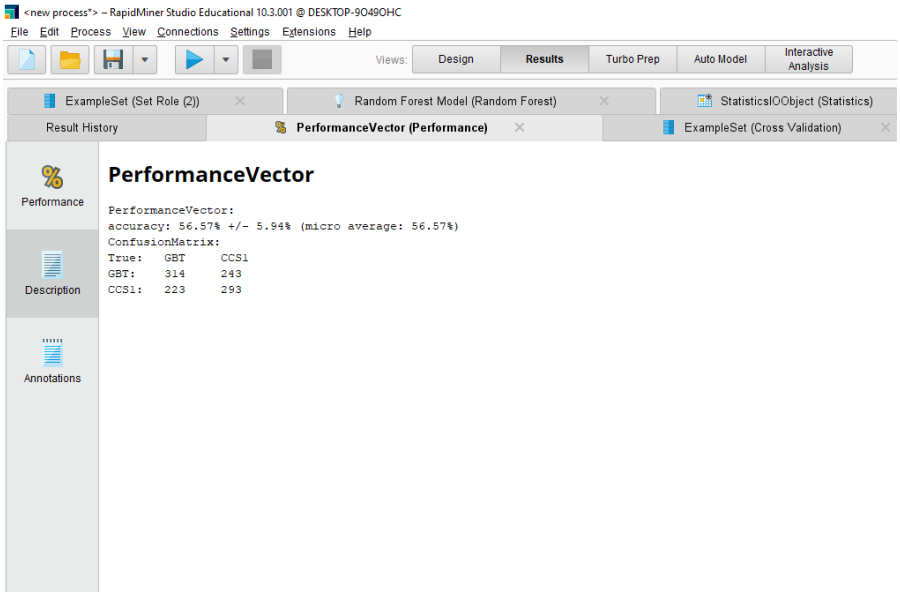


Figure 21: Random forest performance

Accuracy:

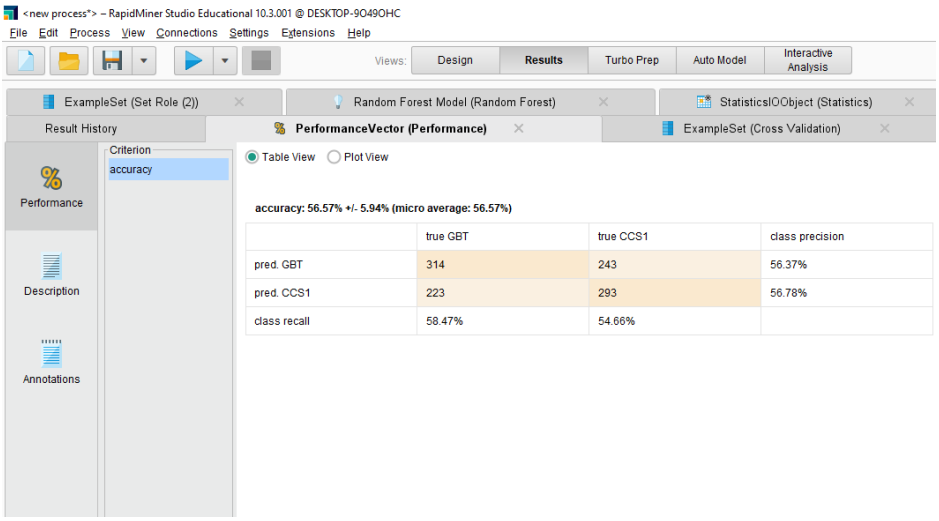


Figure 22: Random forest accuracy

Naïve bayes:

This is a probabilistic classification algorithm based on “Bayes” theorem, assuming independence between features. Its widely used in text classification, spam filtering, and recommendation systems due to its simplicity.

Process:

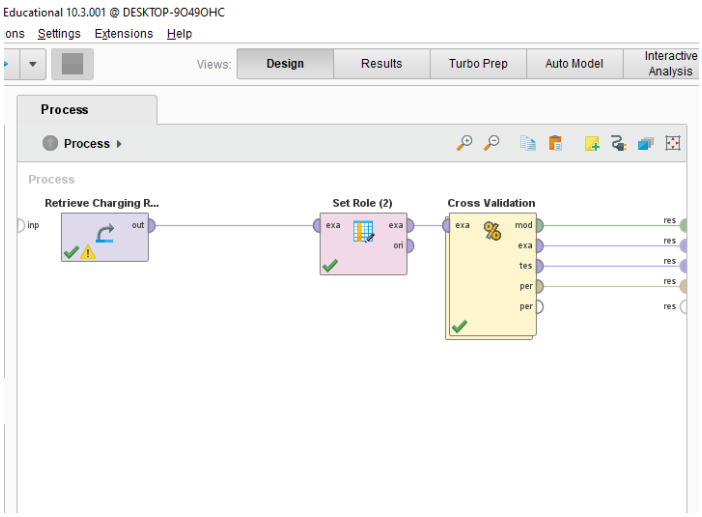


Figure 23: Naive bayes model 1

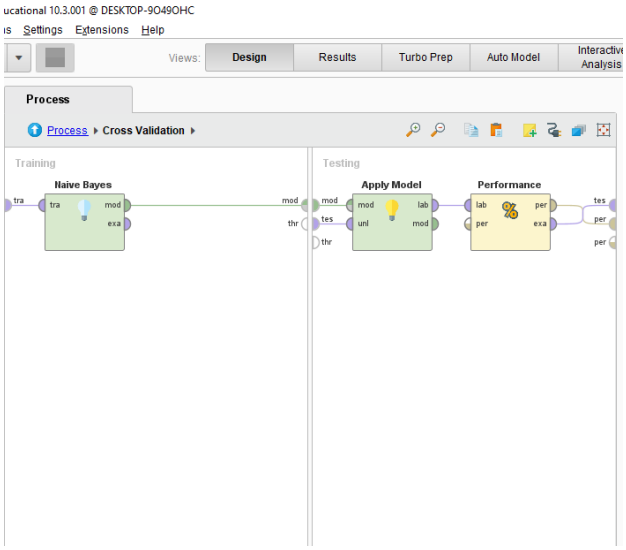
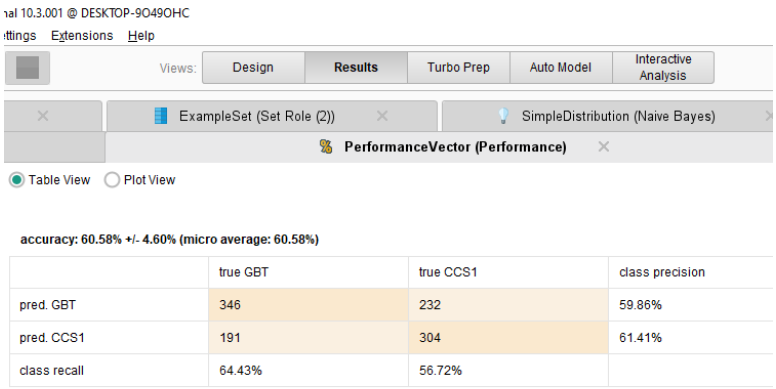


Figure 24: Naïve bayes model 2.

Results:

Connector	prediction(Connector)	confidence(GBT)	confidence(CCS1)
GBT	GBT	0.649	0.351
GBT	GBT	0.702	0.298
GBT	GBT	0.655	0.345
GBT	GBT	0.705	0.295
GBT	CCS1	0.394	0.606
GBT	GBT	0.711	0.289
GBT	GBT	0.727	0.273
GBT	GBT	0.746	0.254
GBT	GBT	0.740	0.260
CCS1	CCS1	0.159	0.841
CCS1	CCS1	0.365	0.635
CCS1	GBT	0.744	0.256
GBT	GBT	0.679	0.321

Figure 25: naive bayes result 1



	true GBT	true CCS1	class precision
pred. GBT	346	232	59.86%
pred. CCS1	191	304	61.41%
class recall	64.43%	56.72%	

Figure 26:Naive bayes result 2

We set a “connector” as a label data to predict what is one the most used to charge the electric cars. So the (GBT) take a 64.43% but (CCS1) take 56.72% these are the true results, but the predictions results (GBT) take 59.86% but (CCS1) take 61.41%, this model predict that (GBT) is better because it can charge Meter Value/KM more than (CCS1).

K-Means:

K-means is a popular unsupervised machine learning algorithm used for clustering tasks. It partitions a dataset into kkk distinct, non-overlapping subsets or clusters based on feature similarity. The goal of K-means is to group data points such that points in the same cluster are more similar to each other than to those in other clusters.

Process:

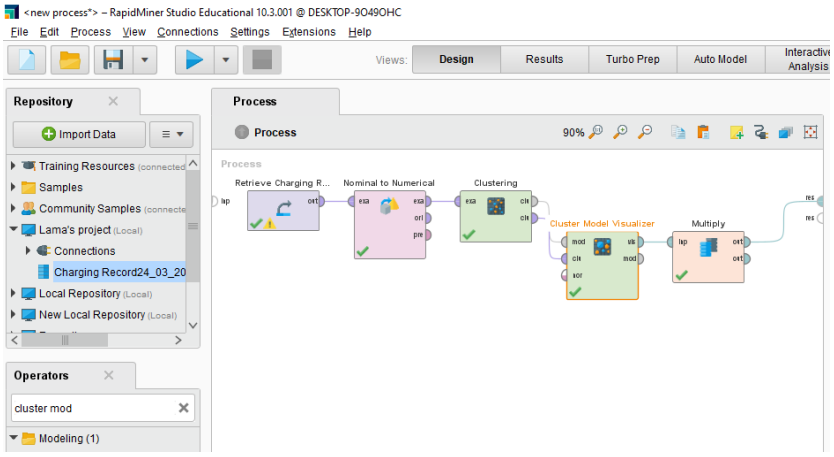


Figure 27: K-Means model

Result:

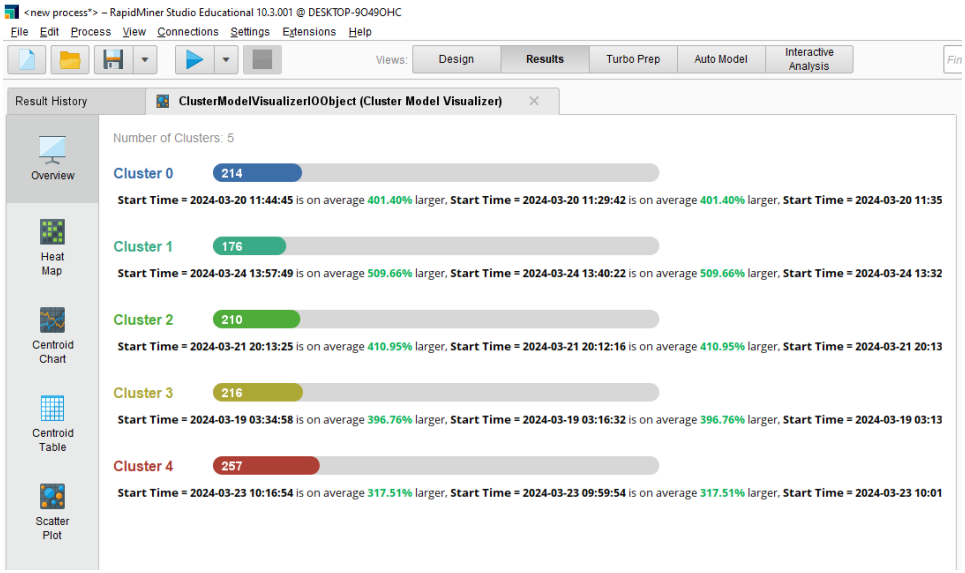


Figure 28: K-Means clustring result 1

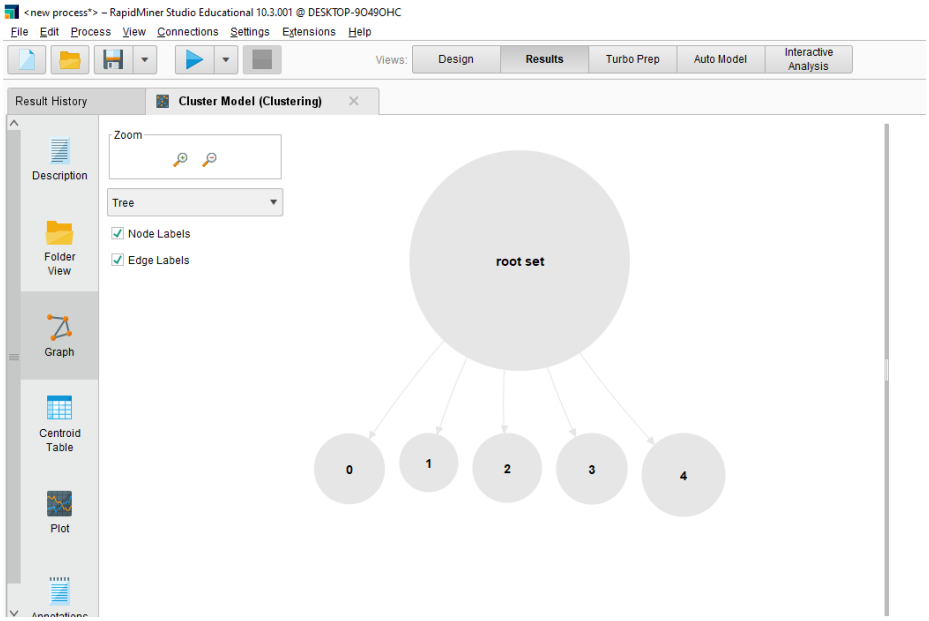


Figure 29: K-Means clustring result 2

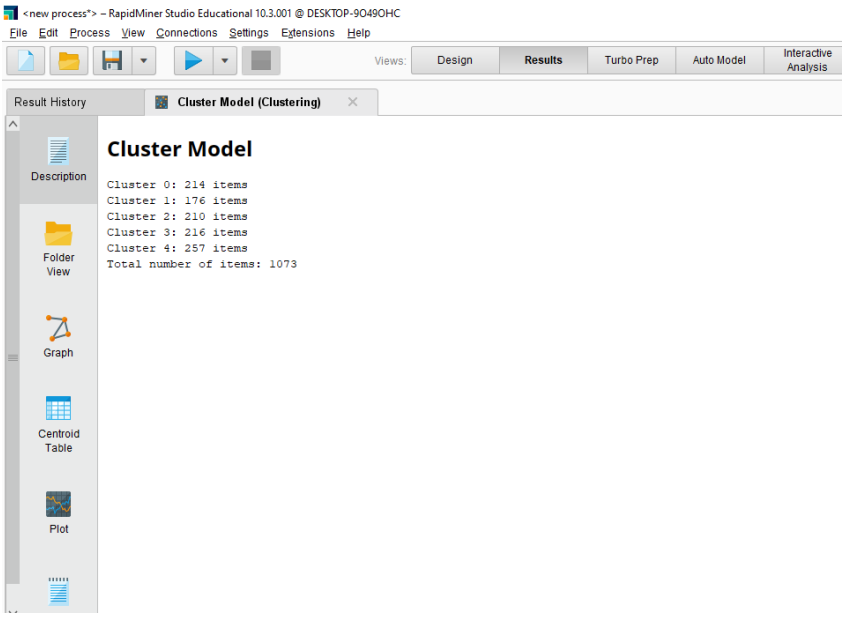


Figure 30: K-Means clustring result 2

Data visualization:

I showcased my data through a Power BI report, incorporating various charts with different fields. The was to ensure that Power BI reports was easily understandable. Here the final reports.

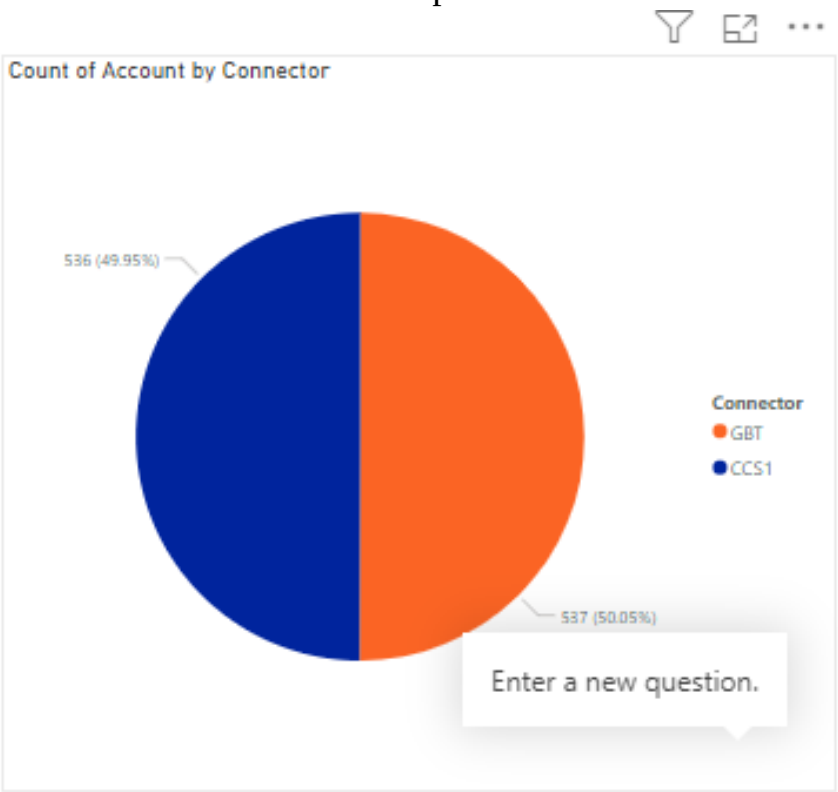


Figure 31: Power BI result 1

As shown in “figure 31” we choose count of account that use the connector type “GBT,CCS1” , referring to the figure we note that the connector “GBT” is used more than “CCS1”.

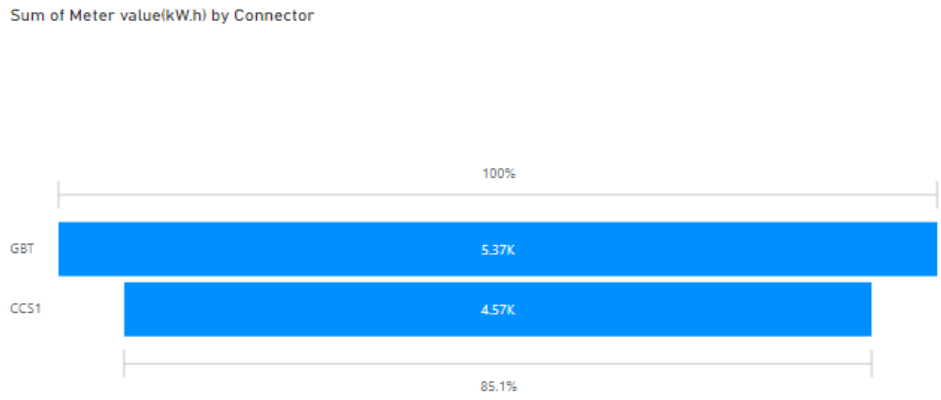


Figure 32:Power BI result 2

As shown in figure"32" we choose the sum of "Meter value(Kw.h) by "Connector type", referring to the figure we note that the "GBT" connector has the ability to charge more than "CCS1" connectore, where the charger "GBT" can charge 5.3K referring to our data

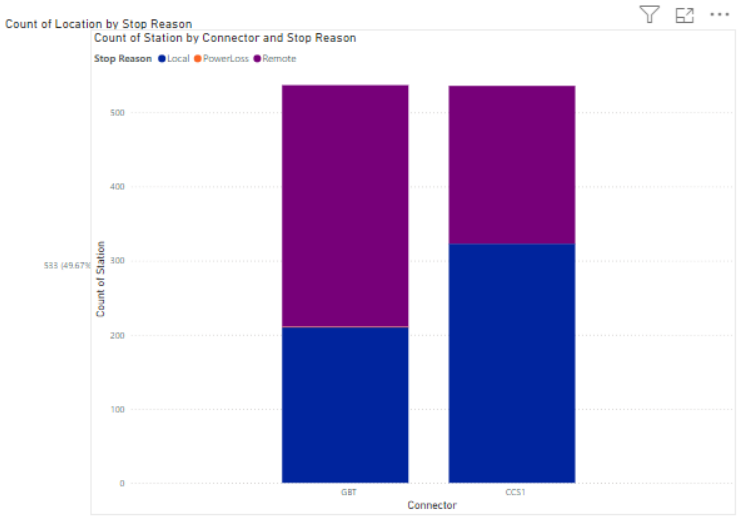


Figure 33:Power BI result 3

In this figure "33" we compared the "connector" type with "stop reason", refrring to the figure and data we can see that most of users enabeled the "GBT"

type remotly that mean by mobile application of charger, but the users of "CCS1" enabled it localy through a card.

Conclusion:

In conclusion, the rapid growth of electric vehicles (EVs) underscores the urgent need for an expansive and efficient network of charging points. This graduation project has thoroughly analyzed the current landscape of EV adoption and identified critical gaps in charging infrastructure. By leveraging data on vehicle usage patterns, urban planning, and energy consumption, we have developed strategic recommendations for the optimal placement and distribution of charging stations. Implementing these recommendations will not only facilitate the seamless integration of EVs into daily life but also contribute significantly to environmental sustainability by reducing reliance on fossil fuels Furthermore, a well-planned charging infrastructure can enhance user convenience, boost consumer confidence in electric mobility, and accelerate the transition towards a greener, more sustainable transportation future. This project highlights the importance of collaborative efforts among government bodies, private sector stakeholders, and the community to achieve a robust and accessible EV charging network.